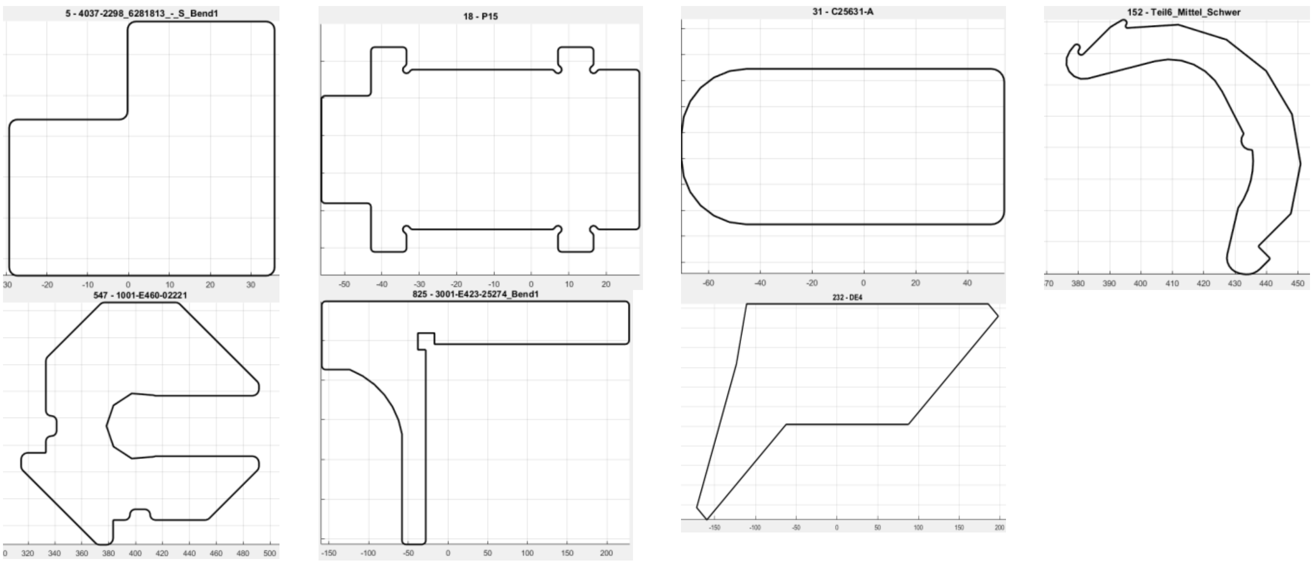
2D Nesting Challenge for CMI

A sheet metal company needs to cut parts out of a large sheet of metal. Therefore they need to arrange all parts on a rectangular metal sheet. Each part need to be cut several times.

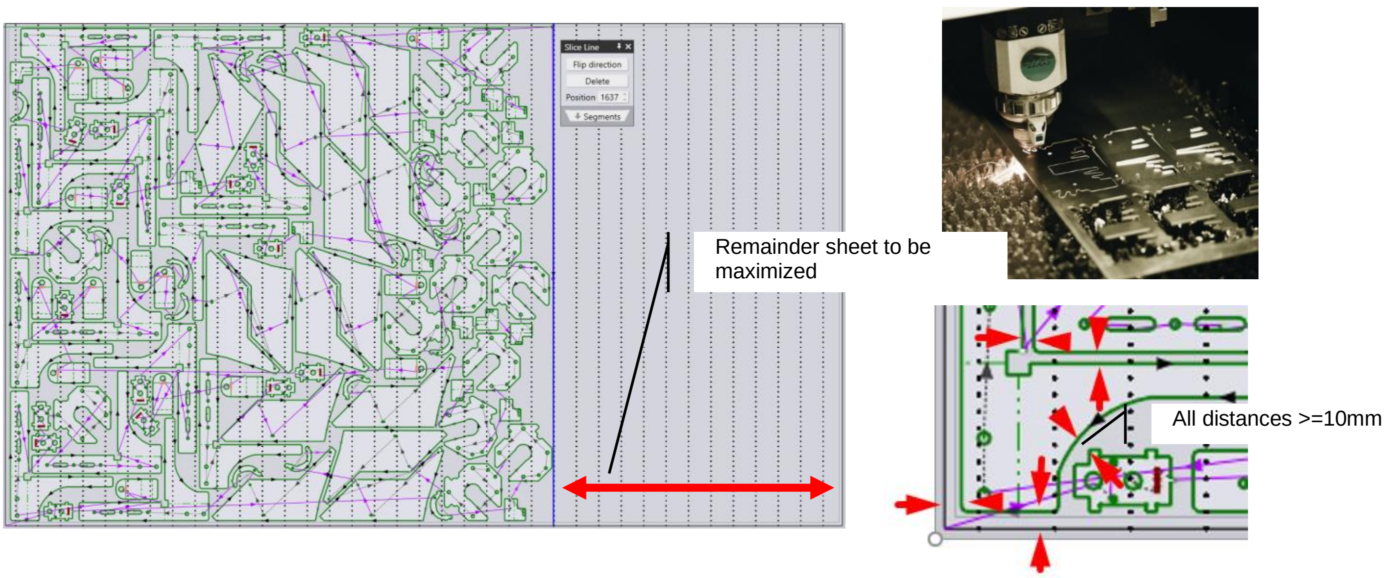
- The size of the raw material is a 2.5m x 1.5m sheet
- Each geometry defined by a polygon.
- For technological reasons the minimal distance between parts and between the parts and the sheet limits must be >=10mm
- The area of the rectangular remainder sheet needs to be maximized (vertical straight cut)

Definition of 2D Nesting: 2D Nesting (referred hereafter simply as “nesting”) is the process of arranging arbitrary 2D shapes inside a bounding rectangular area with some technological constraints.

The following 7 geometries need to be cut in an unknown number . For simplification only the outer contour is given (note: Curves are represented as piecewise linear segments of small length):



Shown below is a simple example of nesting (as an example each part 15 times cut).



Rules

The Challenge:

- Develop and implement an efficient nesting algorithm. Use as minimal material as possible and to maximize overall efficiency. The program needs to terminate within 15 minutes.

Given:

- A work order with a random quantity of the given parts (0-20 times)
- The size of one 2.5x1.5m sheet not to be exceeded

Teams

- Each competitive entry can be either by an individual or a team.
- You may use any programming language of your choice provided it runs on Windows.
- The implementation of the algorithm must be your own work. You must not use any 3rd party code or library that implements nesting.

Files to be submitted to qualify for the final challenge:

- Each student/group needs to create a GitHub repository containing their source code, documentation and a file with instructions on how to run their code.
- A output file with coordinates indicating where cuts will be made
- One file with and example result with each part nested 15 times on the sheet (graphical representation as well as list of points defining the vertices of a polygons).

Evaluation:

- All contestants should provide the following deliverables:
  - A good documentation of the nesting algorithm.
  - Source code of the implementation.
  - An output file that captures the result of the example amount of 15 parts each of the nesting
    - a part manifest specifying for each part, the part name, the position (vertex closest to the origin) and rotation (in relation to the definition provided in the part drawing).
    - A csv list of points defining the vertices of a polygons
    - A graphical visualization of the nest.

The final challenge: "Race day"

- We announce a random amount (1-20) of each part to be nested.
- You have 3 days submit the result of your algorithm in a reproducible way
- We will celebrate all participants and the winners with the largest rectangular remainder sheet with some food and fun at Metamation. It will be a memorable event for everyone 😊

Deadlines:

- Announcement of the part numbers on 15th of April final submission on 18th of April

Awards:

The highest efficiency program will be awarded with 10 000Rs

Contact:

If you have any questions contact careers@metamation.com or stop by at Metamation for a chat

Note: It is a voluntary contest without any claims on the IP by any party or any legal claims on awards. It should give you insight on industrial software design an support your coding skills.

Annexure 1 – File format of a Part

1. Each part is described in a separate text file.
2. A part is defined as a list of points defining the vertices of a closed polygon.
3. Each point is listed on a separate line in terms of its x and y coordinate, separated by a space.

Part 1	Part 2	Part 3	Part 4	Part 5	Part 6	Part 7
17.74 88.08	-56.14 -3.68	1.88 34.51	393.74 24.79	333.24 -15.98	-28.16 188.11	87.65 -78.17
33.74 88.08	-56.14 9.67	49.18 34.51	383.04 22.04	333.24 2.34	-28.16 -51.89	-62.14 -78.17
34.26 88.01	-56.10 9.93	50.47 34.34	380.80 21.93	333.28 2.99	-28.33 -53.18	-62.21 -78.17
34.74 87.81	-56.00 10.17	51.68 33.84	378.70 22.73	333.41 3.63	-28.83 -54.39	-62.27 -78.18
35.16 87.49	-55.85 10.38	52.71 33.05	377.10 24.30	333.62 4.25	-29.63 -55.42	-62.33 -78.20
35.47 87.08	-55.64 10.54	53.51 32.01	376.25 26.38	333.91 4.84	-30.66 -56.22	-62.39 -78.23
35.67 86.60	-55.40 10.64	54.01 30.80	376.31 28.62	334.27 5.38	-31.87 -56.72	-62.45 -78.27
35.74 86.08	-55.14 10.67	54.18 29.51	377.26 30.66	334.70 5.87	-33.16 -56.89	-62.49 -78.31
35.74 27.84	-43.89 10.67	54.18 -20.49	377.26 30.66	373.49 44.66	-53.06 -56.89	-159.21 -175.03
35.67 27.32	-43.63 10.71	54.01 -21.78	377.50 30.98	373.98 45.09	-54.35 -56.72	-159.32 -175.11
35.47 26.84	-43.39 10.81	53.51 -22.99	377.75 31.31	374.52 45.45	-55.56 -56.22	-159.44 -175.16
35.16 26.42	-43.18 10.96	52.71 -24.02	378.00 31.63	375.11 45.74	-56.60 -55.42	-159.57 -175.18
34.74 26.11	-43.02 11.17	51.68 -24.82	378.25 31.95	375.73 45.95	-57.39 -54.39	-159.69 -175.16
34.26 25.91	-42.92 11.41	50.47 -25.32	378.50 32.27	376.37 46.08	-57.89 -53.18	-159.82 -175.11
33.74 25.84	-42.89 11.67	49.18 -25.49	378.75 32.59	377.02 46.12	-58.06 -51.89	-159.92 -175.03
-27.26 25.84	-42.89 22.67	-45.42 -25.49	378.75 32.59	429.45 46.12	-58.06 81.96	-171.85 -163.10
-27.78 25.91	-42.85 22.93	-51.89 -24.64	379.18 32.91	430.10 46.08	-58.06 81.98	-171.91 -163.03
-28.26 26.11	-42.75 23.17	-57.92 -22.14	379.72 32.95	430.74 45.95	-58.06 81.99	-171.95 -162.95
-28.67 26.42	-42.60 23.38	-63.10 -18.17	380.20 32.72	431.36 45.74	-58.06 82.00	-171.98 -162.86
-28.99 26.84	-42.39 23.54	-67.07 -12.99	380.49 32.27	431.95 45.45	-58.06 82.02	-171.99 -162.77
-29.19 27.32	-42.15 23.64	-69.57 -6.96	380.50 31.73	432.49 45.09	-58.07 82.03	-171.99 -162.68
-29.26 27.84	-41.89 23.67	-70.42 -0.49	380.24 31.26	432.99 44.66	-58.07 82.05	-171.97 -162.59
-29.26 62.08	-34.39 23.67	-70.42 9.51	380.24 31.26	490.16 -12.51	-58.07 82.05	-123.34 -16.67
-29.19 62.60	-34.13 23.64	-69.57 15.98	379.98 30.82	490.59 -13.01	-62.55 99.68	-123.33 -16.66
-28.99 63.08	-33.89 23.54	-67.07 22.01	379.98 30.30	490.96 -13.55	-70.01 116.28	-123.33 -16.65
-28.67 63.49	-33.68 23.38	-63.10 27.19	380.24 29.86	491.24 -14.14	-80.23 131.34	-123.33 -16.65
-28.26 63.81	-33.52 23.17	-57.92 31.16	380.69 29.60	491.46 -14.76	-92.89 144.41	-123.33 -16.64
-27.78 64.01	-33.42 22.93	-51.89 33.66	381.20 29.60	491.58 -15.40	-107.62 155.10	-123.32 -16.63
-27.26 64.08	-33.39 22.67	-45.42 34.51	381.65 29.86	491.63 -16.05	-123.97 163.08	-123.32 -16.62
-2.26 64.08	-33.39 19.40	1.88 34.51	390.13 38.34	491.62 -18.30	-123.97 163.08	-110.91 44.22
-1.74 64.15	-33.40 19.23		390.14 38.35	491.45 -19.59	-124.00 163.09	-110.87 44.33
-1.26 64.35	-33.45 19.06		390.15 38.36	490.95 -20.80	-124.03 163.10	-110.82 44.42
-0.84 64.67	-33.52 18.90		390.17 38.38	490.16 -21.84	-124.06 163.10	-110.74 44.50
-0.53 65.08	-33.62 18.76		390.18 38.39	489.12 -22.63	-124.09 163.11	-110.64 44.57
-0.33 65.56	-33.75 18.64		390.20 38.40	487.92 -23.13	-124.12 163.11	-110.53 44.60
-0.26 66.08	-33.89 18.54		390.21 38.41	486.62 -23.30	-124.15 163.11	-110.42 44.62
-0.26 86.08	-33.89 18.54		390.21 38.41	415.23 -23.30	-154.06 163.11	185.23 44.62
-0.19 86.60	-34.29 18.09		390.83 38.80	414.88 -23.29	-155.36 163.28	185.29 44.61
0.01 87.08	-34.37 17.50		391.45 39.18	414.54 -23.25	-156.56 163.78	185.36 44.60
0.33 87.49	-34.10 16.96		392.08 39.56	414.21 -23.19	-157.60 164.58	185.42 44.58
0.74 87.81	-33.56 16.69		392.70 39.94	413.87 -23.11	-158.39 165.61	185.48 44.55
1.22 88.01	-32.97 16.77		393.34 40.30	413.55 -23.01	-158.89 166.82	185.53 44.51
1.74 88.08	-32.52 17.17		393.97 40.67	413.23 -22.88	-159.06 168.11	185.58 44.47
17.74 88.08	-32.42 17.31		393.97 40.67	413.23 -22.88	-159.06 244.03	197.58 32.47
	-32.30 17.44		394.41 40.79	397.06 -21.57	-158.89 245.32	197.66 32.37
	-32.16 17.54		394.87 40.71	383.50 -30.46	-158.39 246.53	197.71 32.25
	-32.00 17.61		395.23 40.43	378.24 -45.80	-157.60 247.56	197.73 32.12
	-31.83 17.66		395.44 40.02	383.50 -61.14	-156.56 248.36	197.71 31.99
	-31.66 17.67		395.44 39.56	397.06 -70.03	-155.36 248.86	197.66 31.87
	5.38 17.67		395.23 39.15	413.23 -68.72	-154.06 249.03	197.58 31.76
	5.55 17.66		395.23 39.15	413.23 -68.72	222.85 249.03	87.65 -78.17
	5.72 17.61		395.11 38.93	413.55 -68.59	224.15 248.86	
	5.88 17.54		395.09 38.68	413.87 -68.49	225.35 248.36	
	6.02 17.44		395.18 38.45	414.21 -68.40	226.39 247.56	
	6.15 17.31		395.35 38.27	414.54 -68.35	227.18 246.53	
	6.25 17.17		395.58 38.17	414.88 -68.31	227.68 245.32	
	6.25 17.17		395.83 38.18	415.23 -68.30	227.85 244.03	
	6.69 16.77		395.83 38.18	486.62 -68.30	227.85 200.03	
	7.29 16.69		411.88 39.21	487.92 -68.47	227.68 198.73	
	7.82 16.96		427.24 34.43	489.12 -68.97	227.18 197.53	
	8.10 17.50		439.87 24.47	490.16 -69.76	226.39 196.49	
	8.02 18.09		448.11 10.66	490.95 -70.80	225.35 195.70	
	7.61 18.54		450.86 -5.19	491.45 -72.00	224.15 195.20	
	7.47 18.64		447.76 -20.97	491.62 -73.30	222.85 195.03	
	7.35 18.76		447.76 -20.97	491.62 -75.55	-16.65 195.03	
	7.25 18.90		447.75 -21.00	491.58 -76.20	-16.78 195.04	
	7.17 19.06		447.73 -21.03	491.45 -76.84	-16.90 195.09	
	7.13 19.23		447.71 -21.06	491.24 -77.46	-17.00 195.17	
	7.11 19.40		447.69 -21.09	490.95 -78.05	-17.08 195.28	
	7.11 22.67		447.67 -21.12	490.59 -78.59	-17.13 195.40	
	7.15 22.93		447.65 -21.14	490.16 -79.08	-17.15 195.53	
	7.25 23.17		437.67 -31.12	454.91 -114.33	-17.15 208.53	
	7.40 23.38		437.59 -31.22	454.42 -114.77	-17.16 208.66	

	7.61 23.54		437.54 -31.34	453.87 -115.13	-17.21 208.78	
	7.85 23.64		437.53 -31.47	453.29 -115.42	-17.29 208.88	
	8.11 23.67		437.54 -31.60	452.67 -115.63	-17.40 208.96	
	15.61 23.67		437.59 -31.72	452.03 -115.76	-17.52 209.01	
	15.87 23.64		437.67 -31.83	451.37 -115.80	-17.65 209.03	
	16.11 23.54		440.86 -35.01	414.74 -115.80	-37.56 209.03	
	16.32 23.38		440.94 -35.11	413.70 -115.66	-37.69 209.01	
	16.48 23.17		440.99 -35.23	412.74 -115.26	-37.81 208.96	
	16.58 22.93		441.00 -35.36	411.91 -114.63	-37.92 208.88	
	16.61 22.67		440.99 -35.49	411.27 -113.80	-37.99 208.78	
	16.61 19.40		440.94 -35.61	410.87 -112.83	-38.04 208.66	
	16.60 19.23		440.86 -35.71	410.74 -111.80	-38.06 208.53	
	16.55 19.06		437.92 -38.65	410.60 -110.76	-38.06 188.61	
	16.48 18.90		437.15 -39.30	410.20 -109.80	-38.04 188.48	
	16.38 18.76		436.28 -39.82	409.57 -108.97	-37.99 188.36	
	16.25 18.64		435.34 -40.18	408.74 -108.33	-37.92 188.26	
	16.11 18.54		434.35 -40.37	407.77 -107.94	-37.81 188.18	
	16.11 18.54		433.34 -40.40	406.74 -107.80	-37.69 188.13	
	15.71 18.09		432.34 -40.26	399.74 -107.80	-37.56 188.11	
	15.63 17.50		431.26 -40.01	398.70 -107.94	-28.16 188.11	
	15.90 16.96		430.03 -39.56	397.74 -108.33		
	16.44 16.69		428.96 -38.80	396.91 -108.97		
	17.03 16.77		428.13 -37.79	396.27 -109.80		
	17.48 17.17		427.59 -36.60	395.87 -110.76		
	17.58 17.31		427.37 -35.31	395.74 -111.80		
	17.70 17.44		427.50 -34.00	395.60 -112.83		
	17.84 17.54		430.93 -19.32	395.20 -113.80		
	18.00 17.61		430.94 -19.29	394.57 -114.63		
	18.17 17.66		430.95 -19.26	393.74 -115.26		
	18.34 17.67		430.96 -19.23	392.77 -115.66		
	27.86 17.67		430.97 -19.20	391.74 -115.80		
	28.12 17.64		430.99 -19.17	383.74 -115.80		
	28.36 17.54		431.01 -19.14	383.61 -115.82		
	28.57 17.38		431.01 -19.14	383.49 -115.87		
	28.73 17.17		432.67 -16.45	383.38 -115.95		
	28.83 16.93		434.00 -13.58	383.30 -116.05		
	28.86 16.67		434.96 -10.56	383.25 -116.17		
	28.86 -24.03		435.54 -7.44	383.24 -116.30		
	28.83 -24.29		435.74 -4.28	383.24 -129.19		
	28.73 -24.53		435.54 -1.12	383.07 -130.48		
	28.57 -24.74		435.54 -1.12	382.57 -131.69		
	28.36 -24.89		435.51 -1.00	381.77 -132.72		
	28.12 -24.99		435.45 -0.90	380.74 -133.52		
	27.86 -25.03		435.37 -0.80	379.53 -134.02		
	18.34 -25.03		435.27 -0.74	378.24 -134.19		
	18.17 -25.01		435.15 -0.70	373.49 -134.19		
	18.00 -24.97		435.03 -0.68	372.84 -134.14		
	17.84 -24.89		435.03 -0.68	372.19 -134.02		
	17.70 -24.79		433.92 -0.51	371.57 -133.81		
	17.58 -24.67		432.96 0.07	370.99 -133.52		
	17.48 -24.53		432.27 0.95	370.44 -133.15		
	17.48 -24.53		431.96 2.03	369.95 -132.72		
	17.03 -24.12		432.06 3.14	316.31 -79.08		
	16.44 -24.04		432.57 4.14	315.88 -78.59		
	15.90 -24.32		432.57 4.14	315.52 -78.05		
	15.63 -24.85		432.62 4.22	315.23 -77.46		
	15.71 -25.45		432.65 4.31	315.02 -76.84		
	16.11 -25.89		432.67 4.40	314.89 -76.20		
	16.25 -25.99		432.67 4.50	314.85 -75.55		
	16.38 -26.12		432.65 4.59	314.85 -70.80		
	16.48 -26.26		432.62 4.68	315.02 -69.50		
	16.55 -26.42		425.87 17.90	315.52 -68.30		
	16.60 -26.59		423.53 21.42	316.31 -67.26		
	16.61 -26.76		420.48 24.33	317.35 -66.47		
	16.61 -30.03		416.84 26.49	318.55 -65.97		
	16.58 -30.29		412.82 27.78	319.85 -65.80		
	16.48 -30.53		408.62 28.14	332.74 -65.80		
	16.32 -30.74		404.43 27.55	332.87 -65.78		
	16.11 -30.89		393.74 24.80	332.99 -65.73		
	15.87 -30.99			333.09 -65.65		
	15.61 -31.03			333.17 -65.55		
	8.11 -31.03			333.22 -65.43		
	7.85 -30.99			333.24 -65.30		
	7.61 -30.89			333.24 -57.30		
	7.40 -30.74			333.37 -56.26		
	7.25 -30.53			333.77 -55.30		
	7.15 -30.29			334.41 -54.47		
	7.11 -30.03			335.24 -53.83		
	7.11 -26.76			336.20 -53.43		
	7.13 -26.59			337.24 -53.30		
	7.17 -26.42			338.27 -53.16		
	7.25 -26.26			339.24 -52.76		
	7.35 -26.12			340.06 -52.13		
	7.47 -25.99			340.70 -51.30		
	7.61 -25.89			341.10 -50.33		
	7.61 -25.89			341.24 -49.30		

	8.02 -25.45 8.10 -24.85 7.82 -24.32 7.29 -24.04 6.69 -24.12 6.25 -24.53 6.15 -24.67 6.02 -24.79 5.88 -24.89 5.72 -24.97 5.55 -25.01 5.38 -25.03 -31.66 -25.03 -31.83 -25.01 -32.00 -24.97 -32.16 -24.89 -32.30 -24.79 -32.42 -24.67 -32.52 -24.53 -32.52 -24.53 -32.97 -24.12 -33.56 -24.04 -34.10 -24.32 -34.37 -24.85 -34.29 -25.45 -33.89 -25.89 -33.75 -25.99 -33.62 -26.12 -33.52 -26.26 -33.45 -26.42 -33.40 -26.59 -33.39 -26.76 -33.39 -30.03 -33.42 -30.29 -33.52 -30.53 -33.68 -30.74 -33.89 -30.89 -34.13 -30.99 -34.39 -31.03 -41.89 -31.03 -42.15 -30.99 -42.39 -30.89 -42.60 -30.74 -42.75 -30.53 -42.85 -30.29 -42.89 -30.03 -42.89 -19.03 -42.92 -18.77 -43.02 -18.53 -43.18 -18.32 -43.39 -18.16 -43.63 -18.06 -43.89 -18.03 -55.14 -18.03 -55.40 -17.99 -55.64 -17.89 -55.85 -17.74 -56.00 -17.53 -56.10 -17.29 -56.14 -17.03 -56.14 -3.68			341.24 -42.30 341.10 -41.26 340.70 -40.30 340.06 -39.47 339.24 -38.83 338.27 -38.43 337.24 -38.30 336.20 -38.16 335.24 -37.76 334.41 -37.13 333.77 -36.30 333.37 -35.33 333.24 -34.30 333.24 -15.98		